

MD. FORHAD BABU

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OBJECTIVE

Final-year undergraduate in Applied Chemistry & Chemical Engineering (ACCE), University of Rajshahi (CGPA 3.65/4.00), seeking a fully-funded Master's or PhD position abroad. Research interests span renewable energy materials (Dye-Sensitized and Perovskite Solar Cells), battery and energy storage systems, biomaterials science, electrochemistry and corrosion science, nanomaterials for wastewater treatment and energy applications, computational chemistry (DFT), and Proficient in Gaussian, GaussView, Material Studio, CAPS1D, MATLAB, and OriginPro, with a strong record of leadership in student research and welfare organizations.

EDUCATION

B.Sc. Engg. in Applied Chemistry & Chemical Engineering (ACCE)

Final Year - Expected Mar 2027

University of Rajshahi, Rajshahi-6205, Bangladesh

- CGPA: 3.65 / 4.00

RESEARCH INTERESTS

- Renewable Energy Materials — Dye-Sensitized Solar Cells (DSSC) and Perovskite Solar Cells (PSC)
- Battery & Energy Storage — electrode materials and electrolyte systems for lithium-ion and beyond-lithium technologies
- Materials Science & Nanomaterials — synthesis and characterization for energy, waste water and biomedical applications
- Electrochemistry & Corrosion — corrosion inhibition and surface characterization
- Computational Chemistry (DFT) — molecular property, reaction mechanism, and electronic structure modeling using Gaussian/GaussView

PUBLICATIONS

1. Hossen, J., **Babu, M. F.**, Dey, T., Siam, S., Islam, M. M., Iqbal, M. M., Rabby, A. M., & Rezvi, M. T. R. (2026). Computational study of the effects of substituent type and position on the O–H bond dissociation energy of phenolic compounds. *Discover Chemistry*, 3, 406. [DOI](#)

CONFERENCE PRESENTATIONS

1. Hossen, J., **Babu, M. F.**, Nakatani, N., Maruf, M. M. R., & Islam, M. S. (2026, May 7–8). Platinum-tetraphenyl Porphyrin Electrocatalyst for Anodic Chlorine Evolution Reaction: Computational DFT Study [**Oral presentation**]. International Conference on Electronics and Informatics 2026 (ICEI 2026), organized by Bangladesh Electronics and Informatics Society, Atomic Energy Centre, Dhaka, Bangladesh. ([Link](#))
2. Hossen, J., Siam, S., **Babu, M. F.**, Dey, T., Islam, M. M., Iqbal, M. M., Rabby, A. M., & Rezvi, M. T. R. (2026, May 7–8). Substituent effects on O–H bond dissociation energy in phenolic antioxidants: A DFT approach [**Oral presentation**, presented by S. Siam]. International Conference on Electronics and Informatics 2026 (ICEI 2026), organized by Bangladesh Electronics and Informatics Society, Atomic Energy Centre, Dhaka, Bangladesh. ([Link](#))
3. Hossen, J., Nakatani, N., **Babu, M. F.**, & Maruf, M. M. R. (2026, April 4–5). Computational investigation of anodic chlorine evolution reaction on platinum-tetraphenyl porphyrin electrocatalyst [**Oral Presentation**]. Presented at the 6th Young Scientist Congress (6th YSC), organized by Bangladesh Academy of Sciences, Dhaka, Bangladesh. Published in *Journal of Bangladesh Academy of Sciences* (2026), 50(Suppl. 1), 43. ([Link](#)) ([Full Volume](#))
4. Islam, M. S., Mia, J., **Babu, M. F.**, Siam, S., Islam, M. M., Dey, T., Tareq, M. J. H., Akter, S., Sarker, P. P., Shahriar, S. M. S., Islam, M. K., & Salam, S. M. A. (2025, December 13). Green corrosion inhibition of steels using Moringa oleifera leaves extract in acidic medium [**Poster presentation**]. Presented at the 1st International Conference on Environmental Sustainability & Greener Earth (ICESGE 2025), organized by American Chemical Society (ACS) International Student Chapter, University of Dhaka, Asiatic Society of Bangladesh, Dhaka, Bangladesh. ([Link](#))

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

1. **President**, July 2026 – Present | **Vice President**, Nov 2025 – July 2026 | **IT Secretary**, March 2025 – Nov 2025 | Rajshahi University Higher Education and Research Forum (RUHERF) | [Link](#)
Spearheaded university-wide research and higher education programs, fostering academic growth.
2. **President**, March 2026 – Present | University Students Association of Sariakandi Sonatola (USASS)
Coordinated initiatives advancing university students' academic, professional, and community development.
3. **National Green Warrior Ambassador**, Dec 2025 - Present | Shishu Ullash Organisation International Global Youth Platform | [Link](#)
Promoted environmental awareness and youth engagement through sustainability initiatives.
4. **Office Secretary**, Oct 2023 – Present | Rajshahi University Students Association of Sariakandi (RUSAS)
Managed official correspondence, documentation, and administrative coordination
5. **IT Secretary**, May 2025 – Apr 2026 | Bogura Zilla Students Welfare Association
Strengthened outreach through digital, IT, and media management.
6. **Treasurer**, Oct 2023 – Present | SWAJON, Shah Makhdum Hall, University of Rajshahi
Maintained financial transparency and promoted blood donation initiatives and awareness.
7. **Active Member**, June 2024 – Present | BD Clean | [Link](#)
Participated in environmental cleanliness and sustainability drives, contributing to eco-awareness campaigns.
8. **Inter Department Relationship Secretary (Ass)**, Sep 2025 – Present | ACCE Group | [Link](#)
Facilitated interdepartmental collaboration and coordinated communication with ACCE Group, Department of Applied Chemistry and Chemical Engineering
9. **Campus Ambassador**, Jan 2026 – Present | BioPc – A Bioinformatics Lab of Research and Training | [Link](#)
Promoted bioinformatics research and training opportunities through campus outreach.

TECHNICAL & SOFT SKILLS

1. **Simulation & Computational Software**
Gaussian, GaussView, CAPS1D, Material Studio, MATLAB
2. **Other Technical Skills**
Scientific Writing, MS Office, OriginPro, Canva Design, Biorender
3. **Soft Skills**
Team Leadership, Research Methodology, Presentation, Problem Solving